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High Performance Computing

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https://bheemhh.github.io/#/HPC26

M1:

- Motivating Parallelism, Scope of Parallel Computing,
- Introduction to HPC:
- Parallel Programming Platforms
  - Implicit Parallelism: Trends in Microprocessor Architectures
  - Limitations of Memory System Performance
- Dichotomy of Parallel Computing Platforms
  - Control Structure of Parallel Platforms
  - Communication Model of Parallel Platforms
- Physical Organization of Parallel Platforms
  - Architecture of an Ideal Parallel Computer
  - Interconnection Networks for Parallel Computers

Explicitly Parallel Platforms

- Dichotomy of Parallel Computing Platforms
  - An **explicitly parallel program** must specify concurrency and interaction between concurrent subtasks.
    - Control structure
    - Parallelism can be expressed at various levels of granularity - from instruction level to processes
- Control Structure of Parallel Programs
  - Processo units either operate under the centralized control of a single control unit or work independently.
  - **Single** control unit that dispatches the same instruction to various processors (that work on different data)

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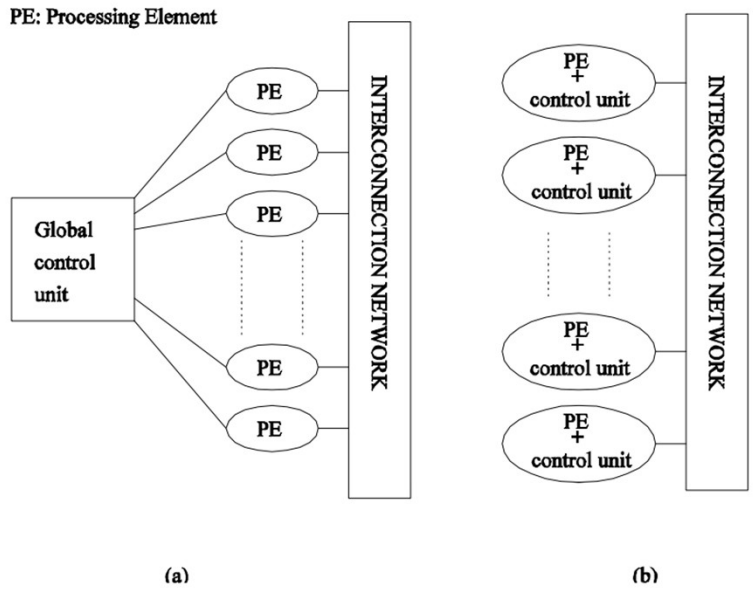
Parallelism

- Classes of parallelism in applications:
  - Data-Level Parallelism (DLP)
  - Task-Level Parallelism (TLP)
- Classes of architectural parallelism:
  - Instruction-Level Parallelism (ILP)
  - Vector architectures/Graphic Processor Units (GPUs)
  - Thread-Level Parallelism
  - Request-Level Parallelism

Control Structure of Parallel Programs

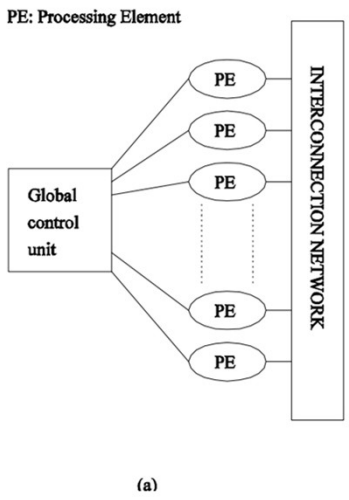
- Single instruction stream, multiple data stream (SIMD)
  - If there is a single control unit that dispatches the same instruction to various processors
- multiple instruction stream, multiple data stream (MiMD)
  - If each processor has its own control control unit, each processor can execute different instructions on different data items



Parallelism from single instruction on multiple processors(SIMD)

- a single control unit dispatches instructions to each processing unit.
- same instruction is executed synchronously by all processing units.
- MMX units in Intel processors and DSP chips
- Structured computations on paral data structures such **as arrays,** oft is necessary to **selectively turn of operations on certain data items**.
- Conditional execution can be detrimental to the performance o SIMD processors and therefore m be used with care.

- Example
  - add instruction is dispatched to all processors and executed concurrently by them.



(a)

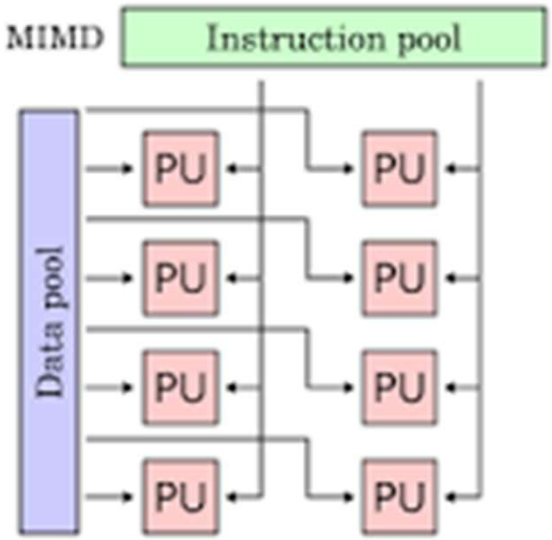
```
for (i = 0; i < 1000; i++)
        c[i] = a[i] + b[i];
```

```
c[0] = a[0] + b[0]; c[1] = a[1] + b[1];,
```

Multiple Instruction stream, Multiple Data stream (MIMD)

- each processing element is capable of executing a different program independent of the other processing elements
- Each of the multiple programs can be inserted into one large if-else block with conditions specified by the task identifiers.
  - A simple variant of this model, called the single program multiple data

    (SPMD) model.
  - An example of MIMD system is **Intel Xeon Phi**,



- SIMD computers require less hardware than MIMD computers because they have only one global control unit.
- SIMD computers require less memory because only one copy of the program needs to be stored.
- While MIMD is complex in terms of complexity than SIMD.
- MIMD is more efficient in terms of performance than SIMD.

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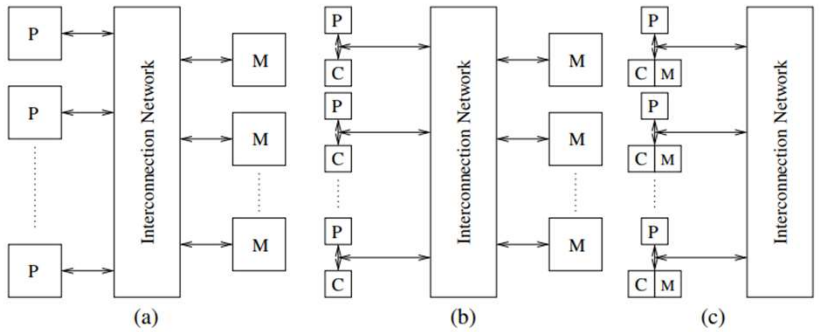
Communication Model of Parallel Platforms

- Data exchange between parallel tasks
  - accessing a shared data space (shared address-space machines or **multiprocessors**)
  - Exchanging messages(message passing platforms or **multicomputers**.
- Shared-Address-Space Platforms
  - common data space that is accessible to all processors.
  - Processors interact by modifying data objects stored in this shared-address-space.
  - If the time taken by a processor to access any memory word in the system global or local is identical, the platform is classified as a uniform memory access (UMA), else, a non uniform memory access (NUMA) machine

NUMA and UMA Shared-Address-Space Platforms

- NUMA and UMA architectures are defined only in terms of **memory access times** and not cache access times.
- Sun Ultra HPC servers-class of NUMA multiprocessors.



Typical shared-address-space architectures: (a) Uniform-memory-access shared-address-space computer; (b Uniform-memory-access shared-address-space computer wit caches and memories; (c) Non-uniform-memory-access shared-address-space computer with local memory only.

NUMA and UMA Shared-Address-Space Platforms

Presence Global Memory (RAM) :

- presence of a global memory space makes (parallel) programming such platforms much easier.
- operations require mutual exclusion for concurrent accesses.
- Read/write -interactions are, however, harder to program than the read-only interactions,

Presence of caches on processors

- raises the issue of multiple copies of a single memory word being manipulated by two or more processors at the same time.
  - Ensuring that **concurrent operations** on multiple copies of the same memory word have well-defined semantics **(cache coherence)**
  - address translation mechanism that locates a memory word in the system.

Message-Passing Platforms

- These platforms comprise of a set of processors and their own (exclusive) memory.
- Instances of such a view come naturally from clustered workstations and non-shared-address-space multicomputers.
- These platforms are programmed using (variants of) send and receive primitives. • Libraries such as MPI and PVM provide such primitives

Message Passing vs. Shared Address Space Platforms

- Message passing requires little hardware support, other than a network.
- Shared address space platforms can easily emulate message passing. The reverse is more difficult to do (in an efficient manner).

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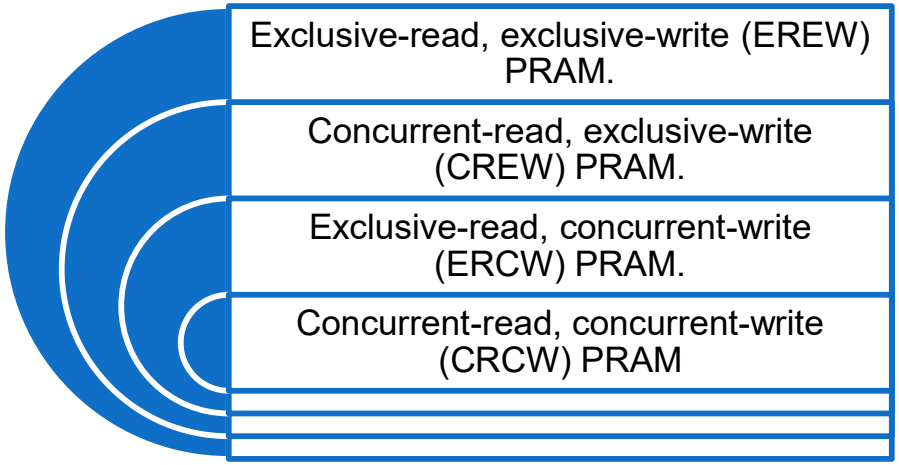
Physical Organization of Parallel Platforms

Architecture of an Ideal Parallel Computer

- parallel random access machine (PRAM)
  - PRAMs consist of p processors and a global memory of unbounded size that is uniformly accessible to all processors.
  - Processors share a common clock but may execute different instructions in each cycle
  - Simultaneous memory accesses are handled, PRAMs can be divided into four subclasses

PRAMs subclasses



concurrent read access does not create any semantic discrepancies

concurrent write access to a memory location requires arbitration

Architecture of an Ideal Parallel Computer

What does concurrent write mean ?

- The most frequently used protocols are as follows:
  - **Common**, in which the concurrent write is allowed if all the values that the processors are attempting to write are identical.
  - **Arbitrary,** in which an arbitrary processor is allowed to proceed with the write operation and the rest fail.
  - **Priority,** in which all processors are organized into a predefined prioritized list, and the processor with the highest priority succeeds and the rest fail.
  - **Sum,** in which the sum of all the quantities is written (the sum-based write conflict resolution model can be extended to any associative operator defined on the quantities being written)

Architectural Complexity of the Ideal Model

- Processors and memories are connected via **switches.**
- System of p processors and m words, the switch complexity is O(mp).
- For a reasonable memory size, constructing a switching network of this complexity is very expensive.
- Thus, PRAM models of computation are impossible to realize in practice
- Solution :
  - **Interconnection Networks** for Parallel Computers
    - A blackbox view of an interconnection network consists **of n inputs and m outputs**

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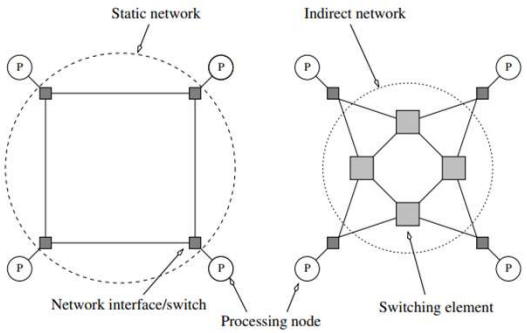
Interconnection Networks for Parallel Computers

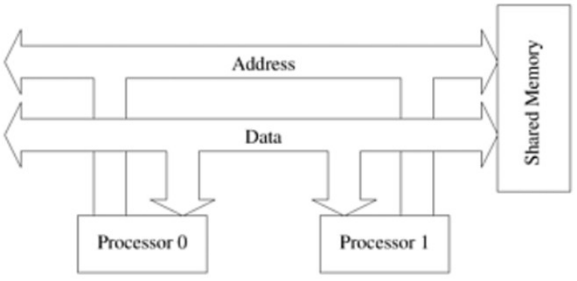
- Interconnects are made of switches and link
- Interconnects are classified as static or dynamic.
- **Static networks** consist of point-to-point communication links among processing nodes and are also referred to as direct networks.
- **Dynamic networks** are built using switches and communication links. Dynamic networks are also referred to as indirect networks



Classification of interconnection networks: (a) a static network; and (b) a dynamic network.

Interconnection Networks Components

- **Switches :**
  - a fixed number of inputs to outputs; Degree of the switch : total number of ports
- **Network Interfaces:**
  - Processors talk to the network via a network interface.
  - connectivity between the nodes and the network
- **Network Topologies:**
  - Bus-Based Networks :
    - Buses are also ideal for broadcasting information among nodes
    - transmission medium is shared, there is little overhead associated with broadcast compared to point-to-point message transfer
    - Reducing shared-bus bandwidth using caches



(a)